

Contact

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Top Skills

OpenAI API

Hugging Face Products

GitHub

Languages

English (Native or Bilingual)

Patents

DATA UPDATES IN DISTRIBUTED
SYSTEM WITH DATA
COHERENCY

Method and system of dynamic
video driver selection on a bootable
CD

Electronically addressed non-volatile
memory-based kernel data cache

Paul Powell

Senior Software Engineer | Systems Engineering | Performance Engineering | Senior Python Engineer and Architect | Available, Reliable, Supportable Software Engineering
Holly Springs, North Carolina, United States

Summary

Well rounded software engineer with experience from high-level application programming (Python / Java / C++) to low-level unix kernel programming (C / C++). Contributed to many very successful projects at leading technology companies. Specializes in storage technologies, such Storage Area Networks and storage protocols.

Experience

NetApp

18 years

Software Engineer - Performance Automation

2017 - 2026 (9 years)

Morrisville, NC

Accelerated environment staging times and eliminated manual setup errors by developing an end-to-end Python automation framework to configure complex NetApp storage systems and client (Linux / Windows) environments.

Optimized performance automation bottlenecks up to 70% using distributed concurrent configuration.

Maximized hardware utilization and streamlined team workflows using a Django web application that efficiently queues and schedules performance runs against constrained, shared laboratory resources.

Expanded test coverage across diverse test requirements by integrating and deploying specialized workload generators designed to stress-test specific storage performance boundaries.

Software Engineer - Web Application

2015 - 2017 (2 years)

Morrisville, NC

Unified disparate support metrics by designing and maintaining a custom PostgreSQL database tailored to ingest and organize complex NetApp customer support data.

Uncovered new up-sell opportunities by developing data-mining algorithms within a Django web application to flag accounts primed for hardware or software upgrades.

Boosted technical sales engagement by engineering an intuitive, social-media-inspired UI dashboard that surfaced critical technical bulletins and high-probability sales leads for account teams.

Software Engineer - Data Protection

2013 - 2016 (3 years)

Morrisville, NC

Delivered a low-RPO data recovery solution by co-designing and implementing an IO synchronous replication system, enabling near-zero data loss capabilities for critical enterprise workloads.

Ensured strict data integrity across distributed environments by programming kernel-level C++ software to replicate granular SCSI protocol commands between remote storage systems.

Improved replication throughput and reduced latency overhead by engineering performance optimizations within the synchronous replication pipeline.

Lead Software Engineer

2008 - April 2013 (5 years)

Research Triangle Park, NC

Guaranteed high availability and reliability for a core SAN configuration component by spearheading the software architecture, kernel-level C programming, and quality control guardrails for a clustered ONTAP management solution.

Accelerated feature delivery by providing technical leadership and architectural mentorship to a software engineering team building multi-threaded, asynchronous software designs.

Eliminated cross-team dependency bottlenecks and optimized project scheduling by managing workloads and spearheading cross-functional collaboration to resolve complex, cluster-wide engineering roadblocks.

Adaptec

Software Engineer

April 2003 - November 2007 (4 years 8 months)

Research Triangle Park, NC

Ensured seamless industry interoperability across Adaptec product lines by serving as the Technical Lead for the implementation of the Storage Management Initiative Specification (SMI-S)

IBM

Software Engineer

January 1999 - April 2003 (4 years 4 months)

Contributed to enterprise-level product stability and feature delivery by writing, debugging, and maintaining high-performance code within a collaborative software engineering team (Java, C++).

Education

Virginia Polytechnic Institute and State University

B.S., Computer Engineering · (1994 - 1998)